

Routing Burp Suite Traffic Through Tor Using FoxyProxy: A Complete Setup Guide

Introduction

Burp Suite is one of the most widely used web security testing tools among penetration testers and security researchers. By default, it acts as an intercepting proxy sitting between your browser and the target application, letting you inspect and manipulate HTTP/HTTPS traffic in real time.

This guide walks through chaining Burp Suite with the Tor network via a SOCKS5 proxy, using FoxyProxy in Firefox to manage the browser-side routing. The result is a layered proxy setup that combines Burp's interception capabilities with Tor's anonymity network.

Traffic flow:

Firefox → FoxyProxy → Burp Suite (127.0.0.1:8080) → Tor SOCKS5 (127.0.0.1:9050) → Internet

⚠ **Scope reminder:** This configuration should only be used for authorized security testing, personal labs, or systems you own or have explicit written permission to assess. Routing traffic through Tor does not grant permission to test systems you don't control.

Prerequisites

- Kali Linux (or similar) with Firefox installed
- Burp Suite (Community or Professional)
- Administrative/sudo access to install packages

Step 1: Configure the Burp Suite Proxy Listener

1. Open Burp Suite.
2. Go to **Proxy → Proxy settings**.
3. Confirm an active listener is bound to:
 - **Bind address:** 127.0.0.1
 - **Port:** 8080

Burp will now accept incoming browser traffic at `http://127.0.0.1:8080`.

Step 2: Install and Configure FoxyProxy in Firefox

FoxyProxy lets Firefox switch quickly between proxy profiles without digging into browser network settings each time.

1. Install the **FoxyProxy** extension from Firefox Add-ons.
2. Open **FoxyProxy → Options → Add New Proxy**.
3. Create a new profile with these settings:

Field	Value
Title	Burp Suite
Proxy Type	HTTP
Host/IP	127.0.0.1
Port	8080

4. Save the profile and enable it from the FoxyProxy toolbar icon.

Firefox traffic now routes through FoxyProxy into Burp Suite.

Step 3: Import the Burp Suite CA Certificate

Since Burp intercepts HTTPS traffic, Firefox needs to trust Burp's Certificate Authority — otherwise you'll hit certificate warnings on every HTTPS site.

1. With the FoxyProxy Burp profile enabled, navigate to `http://burpsuite` in Firefox.
2. Download the **CA Certificate** (typically saved as `cacert.der`).
3. In Firefox, go to **Settings → Privacy & Security → Certificates → View Certificates**.
4. Under the **Authorities** tab, click **Import** and select `cacert.der`.
5. Check **Trust this CA to identify websites**, then click **OK**.

Burp can now decrypt and inspect HTTPS traffic without triggering browser warnings.

Step 4: Install and Start Tor

On Kali Linux:

```
sudo apt update
```

```
sudo apt install tor
```

Start and enable the Tor service:

```
sudo systemctl start tor
```

```
sudo systemctl enable tor
```

```
sudo systemctl status tor
```

Confirm the SOCKS proxy is listening on the default port:

```
ss -lntp | grep 9050
```

Expected output:

```
127.0.0.1:9050
```

Port 9050 is Tor's default SOCKS5 proxy port.

Step 5: Point Burp Suite's Outbound Connections to Tor

1. In Burp Suite, go to **User options** → **Connections**.
2. Find the **SOCKS Proxy** section and enable:
 - ☒ **Use SOCKS proxy**
3. Configure:
 - **SOCKS proxy host:** 127.0.0.1
 - **SOCKS proxy port:** 9050
 - Leave **Username** and **Password** blank.
4. Enable:
 - ☒ **Do DNS lookups over SOCKS proxy** (prevents DNS leaks outside the Tor circuit)

Burp will now forward all outbound connections through Tor. The complete chain:

Firefox → FoxyProxy → Burp Suite (:8080) → SOCKS5 (:9050) → Tor Network → Destination Server

Step 6: Verify the Tor Connection at the Command Line

```
torsocks curl https://check.torproject.org/api/ip
```

Expected response:

```
{
  "IsTor": true,
  "IP": "Tor_exit_node_IP"
}
```

"IsTor": true confirms outbound traffic is exiting through the Tor network.

Step 7: Verify Through Firefox + Burp + Tor

With the FoxyProxy Burp profile still enabled, browse to:

<https://check.torproject.org/>

A correctly configured chain will display:

Congratulations. This browser is configured to use Tor.

The IP shown should be a Tor exit node — not your actual public IP address.

Troubleshooting

Problem: The Tor check shows your real IP address

Work through these checks in order:

1. **Is Burp's SOCKS proxy enabled?** Go to **User options → Connections → SOCKS Proxy** and confirm **Use SOCKS proxy** is checked.
2. **Is the Tor service running?**
3. `sudo systemctl status tor`
4. **Is port 9050 actually listening?**
5. `ss -lntp | grep 9050`
6. **Is FoxyProxy pointed at the right port?** FoxyProxy should target 127.0.0.1:8080 (Burp), **not** 127.0.0.1:9050 (Tor's SOCKS port). Port 9050 belongs exclusively to Burp's own SOCKS configuration — Firefox should never connect to it directly.

Conclusion

Chaining FoxyProxy, Burp Suite, and Tor creates a controlled, layered proxy setup:

Firefox → FoxyProxy → Burp Suite → Tor → Internet

This configuration is a useful learning exercise for topics such as web application security testing, proxy chain analysis, HTTPS interception, traffic inspection, and privacy-focused testing environments.

Use this setup responsibly, and only on systems and networks you own or are explicitly authorized to test.